Proof of predefined time stability of the controller

Step 1. Define $L_1 = (1/2)\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1 + (1/2)\sum_{i=1}^{n}(||\tilde{\boldsymbol{H}}_{i1}||^4)$.
Case 1. $|e_{i,1}| \geq d_{i,1}$

$$\begin{aligned}
\dot{L}_1 = &\sum_{i=1}^{n}\left\{e_{i,1}\left[-\left(\eta_{i,1}+\frac{\beta_1+||\hat{\boldsymbol{H}}_{i1}||^2}{2}\right)\Phi(e_{i,1})-\frac{|\bar{\gamma}_{i,1}|^2\lceil e_{i,1}\rfloor}{|\bar{\gamma}_{i,1}e_{i,1}|+e_1^*}+D_{i,1}-\dot{\chi}_{i,r}\right]\right.\\
&\left.-\|\tilde{\boldsymbol{H}}_{i1}\|^2\left(-\ell_{i,1}\|\hat{\boldsymbol{H}}_{i1}\|^2+\frac{|e_{i,1}|}{2}\right)\right\}+\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_2-\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1\\
\leq &\sum_{i=1}^{n}\left\{-\left(\eta_{i,1}-\frac{||\tilde{\boldsymbol{H}}_{i1}||^2}{2}\right)|e_{i,1}|-\frac{|\bar{\gamma}_{i,1}e_{i,1}|^2}{|\bar{\gamma}_{i,1}e_{i,1}|+e_1^*}+\ell_{i,1}\|\tilde{\boldsymbol{H}}_{i1}\|^4+\frac{\ell_{i,1}\|\boldsymbol{H}_{i1}\|^2}{4}-\ell_{i,1}\|\tilde{\boldsymbol{H}}_{i1}\|^4-\frac{||\tilde{\boldsymbol{H}}_{i1}||^2|e_{i,1}|}{2}\right\}\\
&+\frac{\boldsymbol{\varepsilon}_1^{\mathrm{T}}\boldsymbol{\varepsilon}_1}{2}+\frac{\boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2}{2}\\
\leq &\sum_{i=1}^{n}\left\{-\left(\eta_{i,1}-\frac{||\tilde{\boldsymbol{H}}_{i1}||^2}{2}\right)|e_{i,1}|-|\bar{\gamma}_{i,1}e_{i,1}|+e_1^*+\frac{\ell_{i,1}\|\boldsymbol{H}_{i1}\|^2}{4}-\frac{||\tilde{\boldsymbol{H}}_{i1}||^2|e_{i,1}|}{2}\right\}+\frac{\boldsymbol{\varepsilon}_1^{\mathrm{T}}\boldsymbol{\varepsilon}_1}{2}+\frac{\boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2}{2}\\
\leq &\sum_{i=1}^{n}\left\{-\eta_{i,1}|e_{i,1}|-\zeta|e_{i,1}|^{2-\kappa}\exp\left(|e_{i,1}|^{\kappa}\right)+e_1^*+\frac{\bar{\varepsilon}_{i,1}^2}{2}+\frac{\ell_{i,1}||\boldsymbol{H}_{i1}||^4}{4}\right\}+\frac{\boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2}{2}
\end{aligned} \tag{1}$$

Case 2. $|e_{i,1}| < d_{i,1}$

$$\begin{aligned}
\dot{L}_1 = &\sum_{i=1}^{n}\left\{e_{i,1}\left[-\left(\eta_{i,1}+\frac{\beta_1+||\hat{\boldsymbol{H}}_{i1}||^2}{2}\right)\Phi(e_{i,1})-\frac{|\bar{\gamma}_{i,1}|^2\lceil e_{i,1}\rfloor}{|\bar{\gamma}_{i,1}e_{i,1}|+e_1^*}+D_{i,1}-\dot{\chi}_{i,r}\right]\right.\\
&\left.-\|\tilde{\boldsymbol{H}}_{i1}\|^2\left(-\ell_{i,1}\|\hat{\boldsymbol{H}}_{i1}\|^2+\frac{|e_{i,1}|}{2}\right)\right\}+\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_2-\boldsymbol{e}_1^{\mathrm{T}}\boldsymbol{e}_1\\
\leq &\sum_{i=1}^{n}\left\{\frac{d_{i,1}\left(\beta_1+||\boldsymbol{H}_{i1}||^2\right)}{2}-|\bar{\gamma}_{i,1}e_{i,1}|+e_1^*+\ell_{i,1}||\tilde{\boldsymbol{H}}_{i1}||^2||\hat{\boldsymbol{H}}_{i1}||^2-\frac{||\tilde{\boldsymbol{H}}_{i1}||^2|e_{i,1}|}{2}\right\}+\frac{\boldsymbol{\varepsilon}_1^{\mathrm{T}}\boldsymbol{\varepsilon}_1}{2}+\frac{\boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2}{2}\\
\leq &\sum_{i=1}^{n}\left\{\frac{d_{i,1}\left(\beta_1+||\boldsymbol{H}_{i1}||^2\right)}{2}-\zeta|e_{i,1}|^{2-\kappa}\exp\left(|e_{i,1}|^{\kappa}\right)+e_1^*+\frac{\ell_{i,1}||\tilde{\boldsymbol{H}}_{i1}||^4}{2}+\frac{\ell_{i,1}||\boldsymbol{H}_{i1}||^4}{2}-\ell_{i,1}||\tilde{\boldsymbol{H}}_{i1}||^4\right.\\
&\left.+\frac{\ell_{i,1}||\tilde{\boldsymbol{H}}_{i1}||^4}{2}+\frac{d_{i,1}^2}{2\ell_{i,1}}\right\}+\frac{\boldsymbol{\varepsilon}_1^{\mathrm{T}}\boldsymbol{\varepsilon}_1}{2}+\frac{\boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2}{2}\\
\leq &\sum_{i=1}^{n}\left\{-\zeta|e_{i,1}|^{2-\kappa}\exp\left(|e_{i,1}|^{\kappa}\right)+e_1^*+\frac{\bar{\varepsilon}_{i,1}^2}{2}+\frac{d_{i,1}\left(\beta_1+||\boldsymbol{H}_{i1}||^2\right)}{2}+\frac{\ell_{i,1}||\boldsymbol{H}_{i1}||^4}{2}+\frac{d_{i,1}^2}{2\ell_{i,1}}\right\}+\frac{\boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2}{2}
\end{aligned} \tag{2}$$

Step 2. Design $L_2 = (1/2)\boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2 + (1/2)\sum_{i=1}^{n}(||\tilde{\boldsymbol{H}}_{i2}||^4)$.

Case 1. $|e_{i,2}| \geq d_{i,2}$

$$\begin{aligned}
\dot{L}_2 &= \sum_{i=1}^{n}\Bigg\{e_{i,2}\left[-\left(\eta_{i,2} + \frac{\beta_2 + ||\hat{\boldsymbol{H}}_{i2}||^2}{2}\right)\Phi(e_{i,2}) + \boldsymbol{H}_{i2}^{\mathrm{T}}\boldsymbol{\psi}_2(\boldsymbol{N}_2) + \varepsilon_{i,2}\right] - \frac{|\bar{\gamma}_{i,2}|^2 e_{i,2}\lceil e_{i,2}\rfloor}{|\bar{\gamma}_{i,2}e_{i,2}| + e_2^*} \\
&\quad - \|\tilde{\boldsymbol{H}}_{i2}\|^2\left(-\ell_{i,2}\|\hat{\boldsymbol{H}}_{i2}\|^2 + \frac{|e_{i,2}|}{2}\right)\Bigg\} + \boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{e}_3 - \frac{3}{2}\boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2 \\
&\leq \sum_{i=1}^{n}\Bigg\{-\left(\eta_{i,2} - \frac{||\tilde{\boldsymbol{H}}_{i2}||^2}{2}\right)|e_{i,2}| - \frac{|\bar{\gamma}_{i,2}e_{i,2}|^2}{|\bar{\gamma}_{i,2}e_{i,2}| + e_2^*} + \ell_{i,2}\|\tilde{\boldsymbol{H}}_{i2}\|^4 + \frac{\ell_{i,2}\|\boldsymbol{H}_{i2}\|^2}{4} - \ell_{i,2}\|\tilde{\boldsymbol{H}}_{i2}\|^4 \\
&\quad - \frac{||\tilde{\boldsymbol{H}}_{i2}||^2|e_{i,2}|}{2}\Bigg\} + \boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{e}_3 + \frac{\boldsymbol{\varepsilon}_2^{\mathrm{T}}\boldsymbol{\varepsilon}_2}{2} - \boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2 \\
&\leq \sum_{i=1}^{n}\left\{-\eta_{i,2}|e_{i,2}| - \zeta|e_{i,2}|^{2-\kappa}\exp\left(|e_{i,2}|^{\kappa}\right) + e_2^* + \frac{\bar{\varepsilon}_{i,2}^2}{2} + \frac{\ell_{i,2}||\boldsymbol{H}_{i2}||^4}{4}\right\} + \frac{\bar{k}^2\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3}{2} - \frac{\boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2}{2}
\end{aligned} \tag{3}$$

$\bar{k}^2$ is defined as greater than$||\boldsymbol{M}^{-1}\boldsymbol{K}||^2$.

Case 2. $|e_{i,2}| < d_{i,2}$

$$\begin{aligned}
\dot{L}_2 &= \sum_{i=1}^{n}\Bigg\{e_{i,2}\left[-\left(\eta_{i,2} + \frac{\beta_2 + ||\hat{\boldsymbol{H}}_{i2}||^2}{2}\right)\Phi(e_{i,2}) + \boldsymbol{H}_{i2}^{\mathrm{T}}\boldsymbol{\psi}_2(\boldsymbol{N}_2) + \varepsilon_{i,2}\right] - \frac{|\bar{\gamma}_{i,2}|^2 e_{i,2}\lceil e_{i,2}\rfloor}{|\bar{\gamma}_{i,2}e_{i,2}| + e_2^*} \\
&\quad - \|\tilde{\boldsymbol{H}}_{i2}\|^2\left(-\ell_{i,2}\|\hat{\boldsymbol{H}}_{i2}\|^2 + \frac{|e_{i,2}|}{2}\right)\Bigg\} + \boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{e}_3 - \frac{3\boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2}{2} \\
&\leq \sum_{i=1}^{n}\left\{\left(\frac{\beta_2 + ||\boldsymbol{H}_{i2}||^2}{2} + \varepsilon_{i,2}\right)|e_{i,2}| - \frac{|\bar{\gamma}_{i,2}e_{i,2}|^2}{|\bar{\gamma}_{i,2}e_{i,2}| + e_2^*} + \ell_{i,2}||\tilde{\boldsymbol{H}}_{i2}||^2||\hat{\boldsymbol{H}}_{i2}||^2 - \frac{||\tilde{\boldsymbol{H}}_{i2}||^2|e_{i,2}|}{2}\right\} \\
&\quad + \boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{M}^{-1}\boldsymbol{K}\boldsymbol{e}_3 - \frac{3\boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2}{2} \\
&\leq \sum_{i=1}^{n}\left\{-\zeta|e_{i,2}|^{2-\kappa}\exp\left(|e_{i,2}|^{\kappa}\right) + \frac{d_{i,2}\left(\beta_2 + ||\boldsymbol{H}_{i2}||^2\right)}{2} + e_2^* + \frac{\bar{\varepsilon}_{i,2}^2}{2} + \frac{\ell_{i,2}||\boldsymbol{H}_{i2}||^2}{2} + \frac{d_{i,2}^2}{2\ell_{i,2}}\right\} \\
&\quad + \frac{\bar{k}^2\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3}{2} - \frac{\boldsymbol{e}_2^{\mathrm{T}}\boldsymbol{e}_2}{2}
\end{aligned} \tag{4}$$

Step 3. Select $L_3 = (1/2)\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3 + (1/2)\sum_{i=1}^{n}(||\tilde{\boldsymbol{H}}_{i3}||^4)$.

Case 1. $|e_{i,3}| \geq d_{i,3}$

$$\begin{aligned}
\dot{L}_3 &= \sum_{i=1}^{n}\Bigg\{e_{i,3}\left[-\left(\eta_{i,3} + \frac{\beta_3 + ||\hat{\boldsymbol{H}}_{i3}||^2}{2}\right)\Phi(e_{i,3}) + \frac{\beta_3 + ||\boldsymbol{H}_{i3}||^2}{2} + \varepsilon_{i,3}\right] - \frac{|\bar{\gamma}_{i,3}|^2 e_{i,3}\lceil e_{i,3}\rfloor}{|\bar{\gamma}_{i,3}e_{i,3}| + e_3^*} \\
&\quad -\|\tilde{\boldsymbol{H}}_{i3}\|^2\left(-\ell_{i,3}\|\hat{\boldsymbol{H}}_{i3}\|^2 + \frac{|e_{i,3}|}{2}\right)\Bigg\} + \boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_4 - \frac{(\bar{k}^2 + 2)\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3}{2} \\
&\leq \sum_{i=1}^{n}\left\{-\eta_{i,3}|e_{i,3}| - \zeta|e_{i,3}|^{2-\kappa}\exp\left(|e_{i,3}|^{\kappa}\right) + e_3^* + \frac{\bar{\varepsilon}_{i,3}^2}{2} + \frac{\ell_{i,3}||\boldsymbol{H}_{i3}||^4}{4}\right\} + \frac{\boldsymbol{e}_4^{\mathrm{T}}\boldsymbol{e}_4}{2} - \frac{\bar{k}^2\boldsymbol{e}_3^{\mathrm{T}}\boldsymbol{e}_3}{2}
\end{aligned} \tag{5}$$

Case 2. If $|e_{i,3}| < d_{i,3}$, it follows that

$$\begin{aligned}
\dot{L}_3 \le & \sum_{i=1}^{n} \left\{ \frac{d_{i,3}(\beta_3 + ||\boldsymbol{H}_{i3}||^2)}{2} + |e_{i,3}|\varepsilon_{i,3} - \frac{|\bar{\gamma}_{i,3}|^2 e_{i,3} \lceil e_{i,3} \rfloor}{|\bar{\gamma}_{i,3} e_{i,3}| + e_3^*} - \|\tilde{\boldsymbol{H}}_{i3}\|^2 \left(-\ell_{i,3} \|\hat{\boldsymbol{H}}_{i3}\|^2 + \frac{|e_{i,3}|}{2} \right) \right\} \\
& + \boldsymbol{e}_3^{\mathrm{T}} \boldsymbol{e}_4 - \frac{(\bar{k}^2 + 2)\boldsymbol{e}_3^{\mathrm{T}} \boldsymbol{e}_3}{2} \\
\le & \sum_{i=1}^{n} \left\{ -\zeta |e_{i,3}|^{2-\kappa} \exp\left(|e_{i,3}|^{\kappa}\right) + \frac{d_{i,3}(\beta_3 + ||\boldsymbol{H}_{i3}||^2)}{2} + e_3^* + \frac{\bar{\varepsilon}_{i,3}^2}{2} + \frac{\ell_{i,3} ||\boldsymbol{H}_{i3}||^4}{2} + \frac{d_{i,3}^2}{2\ell_{i,3}} \right\} \\
& + \frac{\boldsymbol{e}_4^{\mathrm{T}} \boldsymbol{e}_4}{2} - \frac{\bar{k}^2 \boldsymbol{e}_3^{\mathrm{T}} \boldsymbol{e}_3}{2}
\end{aligned} \tag{6}$$

Step 4. Choose $L_4 = (1/2)\boldsymbol{e}_4^{\mathrm{T}}\boldsymbol{e}_4 + (1/2)\sum_{i=1}^{n}(||\tilde{\boldsymbol{H}}_{i4}||^4)$.
Case 1. If $|e_{i,4}| \ge d_{i,4}$, $\dot{L}_4$ is calculated as

$$\begin{aligned}
\dot{L}_4 = & \sum_{i=1}^{n} \left\{ e_{i,4} \left[-\left(\eta_{i,4} + \frac{\beta_4 + ||\hat{\boldsymbol{H}}_{i4}||^2}{2} \right) \Phi(e_{i,4}) + \boldsymbol{H}_{i4}^{\mathrm{T}} \boldsymbol{\psi}_4(\boldsymbol{N}_4) + \varepsilon_{i,4} - \zeta \lceil e_{i,4} \rfloor^{1-\kappa} \exp(|e_{i,4}|)^{\kappa} \right] \right. \\
& \left. - \|\tilde{\boldsymbol{H}}_{i4}\|^2 \left(-\ell_{i,4} \|\hat{\boldsymbol{H}}_{i4}\|^2 + \frac{|e_{i,4}|}{2} \right) \right\} - \boldsymbol{e}_4^{\mathrm{T}} \boldsymbol{e}_4 + \boldsymbol{e}_4^{\mathrm{T}} (\boldsymbol{J}^{-1} \boldsymbol{R}(\boldsymbol{\xi}) (\boldsymbol{J}^{-1})^{\mathrm{T}} \boldsymbol{N}(\boldsymbol{\varrho}) \boldsymbol{\tau}_v - \boldsymbol{\tau}_v) \\
\le & \sum_{i=1}^{n} \left\{ -\left(\eta_{i,4} - \frac{||\tilde{\boldsymbol{H}}_{i4}||^2}{2} \right) |e_{i,4}| - \zeta |e_{i,4}|^{2-\kappa} \exp(|e_{i,4}|)^{\kappa} + \ell_{i,4} \|\tilde{\boldsymbol{H}}_{i4}\|^4 + \frac{\ell_{i,4} \|\boldsymbol{H}_{i4}\|^2}{4} - \ell_{i,4} \|\tilde{\boldsymbol{H}}_{i4}\|^4 \right. \\
& \left. - \frac{||\tilde{\boldsymbol{H}}_{i4}||^2 |e_{i,4}|}{2} \right\} - \frac{\boldsymbol{e}_4^{\mathrm{T}} \boldsymbol{e}_4}{2} + \frac{\boldsymbol{\varepsilon}_4^{\mathrm{T}} \boldsymbol{\varepsilon}_4}{2} + \boldsymbol{e}_4^{\mathrm{T}} (\boldsymbol{F}(\boldsymbol{\xi}) \boldsymbol{N}(\boldsymbol{\varrho}) \boldsymbol{\tau}_v - \boldsymbol{\tau}_v) \\
\le & \sum_{i=1}^{n} \left\{ -\eta_{i,4} |e_{i,4}| - \zeta |e_{i,4}|^{2-\kappa} \exp(|e_{i,4}|)^{\kappa} + \frac{\bar{\varepsilon}_{i,4}^2}{2} + \frac{\ell_{i,4} ||\boldsymbol{H}_{i4}||^4}{4} + \frac{1}{g_i}(-F_i(t) N(\varrho_i) + 1)\dot{\varrho}_i \right\} - \frac{\boldsymbol{e}_4^{\mathrm{T}} \boldsymbol{e}_4}{2}
\end{aligned} \tag{7}$$

Case 2. When $|e_{i,4}| < d_{i,4}$, $\dot{L}_4$ is given by

$$\begin{aligned}
\dot{L}_4 = & \sum_{i=1}^{n} \left\{ e_{i,4} \left[-\left(\eta_{i,4} + \frac{\beta_4 + ||\hat{\boldsymbol{H}}_{i4}||^2}{2} \right) \Phi(e_{i,4}) + \boldsymbol{H}_{i4}^{\mathrm{T}} \boldsymbol{\psi}_4(\boldsymbol{N}_4) + \varepsilon_{i,4} - \zeta \lceil e_{i,4} \rfloor^{1-\kappa} \exp(|e_{i,4}|)^{\kappa} \right] \right. \\
& \left. - \|\tilde{\boldsymbol{H}}_{i4}\|^2 \left(-\ell_{i,4} \|\hat{\boldsymbol{H}}_{i4}\|^2 + \frac{|e_{i,4}|}{2} \right) \right\} - \boldsymbol{e}_4^{\mathrm{T}} \boldsymbol{e}_4 + \boldsymbol{e}_4^{\mathrm{T}} (\boldsymbol{J}^{-1} \boldsymbol{R}(\boldsymbol{\xi}) (\boldsymbol{J}^{-1})^{\mathrm{T}} \boldsymbol{N}(\boldsymbol{\varrho}) \boldsymbol{\tau}_v - \boldsymbol{\tau}_v) \\
\le & \sum_{i=1}^{n} \left\{ \frac{d_{i,4}(\beta_4 + ||\boldsymbol{H}_{i4}||^2)}{2} - \zeta |e_{i,4}|^{2-\kappa} \exp(|e_{i,4}|)^{\kappa} + \ell_{i,4} ||\tilde{\boldsymbol{H}}_{i4}||^2 ||\hat{\boldsymbol{H}}_{i4}||^2 - \frac{||\tilde{\boldsymbol{H}}_{i4}||^2 |e_{i,4}|}{2} \right\} \\
& - \frac{\boldsymbol{e}_4^{\mathrm{T}} \boldsymbol{e}_4}{2} + \frac{\boldsymbol{\varepsilon}_4^{\mathrm{T}} \boldsymbol{\varepsilon}_4}{2} + \boldsymbol{e}_4^{\mathrm{T}} (\boldsymbol{F}(\boldsymbol{\xi}) \boldsymbol{N}(\boldsymbol{\varrho}) \boldsymbol{\tau}_v - \boldsymbol{\tau}_v) \\
\le & \sum_{i=1}^{n} \left\{ -\zeta |e_{i,4}|^{2-\kappa} \exp(|e_{i,4}|)^{\kappa} + \frac{d_{i,4}(\beta_4 + ||\boldsymbol{H}_{i4}||^2)}{2} + \frac{\bar{\varepsilon}_{i,4}^2}{2} + \frac{\ell_{i,4} ||\boldsymbol{H}_{i4}||^4}{2} + \frac{d_{i,4}^2}{2\ell_{i,4}} \right. \\
& \left. + \frac{1}{g_i}(-F_i(t) N(\varrho_i) + 1)\dot{\varrho}_i \right\} - \frac{\boldsymbol{e}_4^{\mathrm{T}} \boldsymbol{e}_4}{2}
\end{aligned} \tag{8}$$

$F_i(t)$ is bounded by some unknown set.